Machine learning for breast cancer diagnosis

Elsa Bidant – September 2024

The code developed for this analysis is available at: https://github.com/elsabidant/ML-for-cancer-diagnosis

I. Introduction

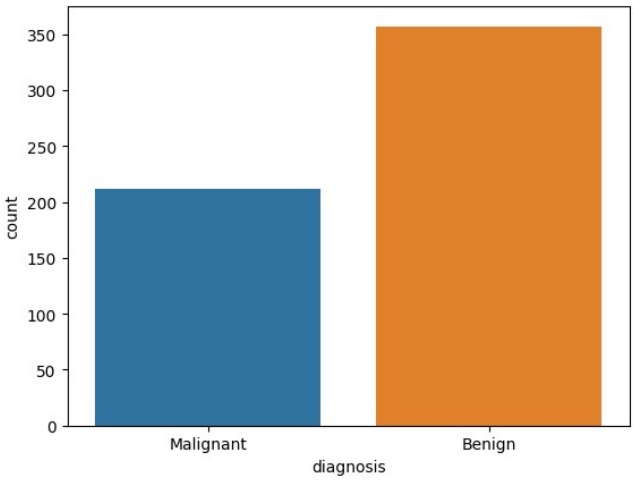
Early and reliable cancer detection remains a central challenge in modern healthcare. Machine learning offers powerful tools to assist clinical decision-making by identifying patterns in structured medical data that may not be immediately visible through traditional statistical approaches.

The objective of this project is to build and evaluate supervised learning models capable of predicting whether a tumour is malignant or benign based on clinical features.

This project aims to answer the following question, to what extent can classical machine learning models accurately classify tumours using structured clinical variables?

II. Data

The dataset used contains clinical and diagnostic measurements extracted from tumour examinations. Each observation corresponds to a patient case and includes quantitative medical features and a binary target variable (malignant / benign).



The final working dataset was split into training and testing sets to ensure proper model validation.

III. Method

The project follows a structured machine learning pipeline. First, the data were standardized and split into training and test sets. Five supervised classification algorithms were then implemented and compared:

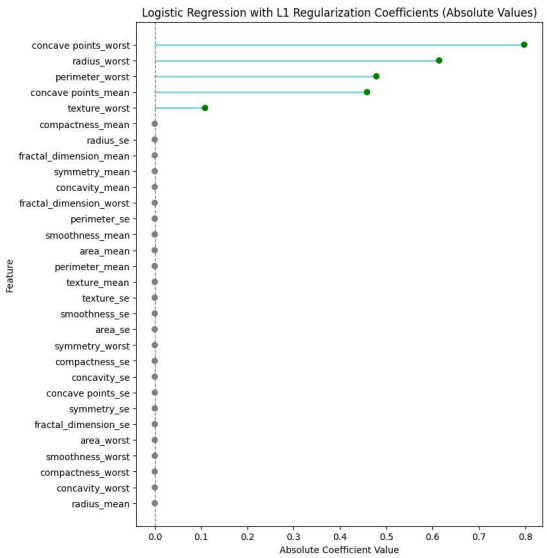
- Logistic regression
- Logistic regression with L1 (Lasso) regularization
- Logistic regression with L2 (Ridge) regularization
- Random forest
- Neural network

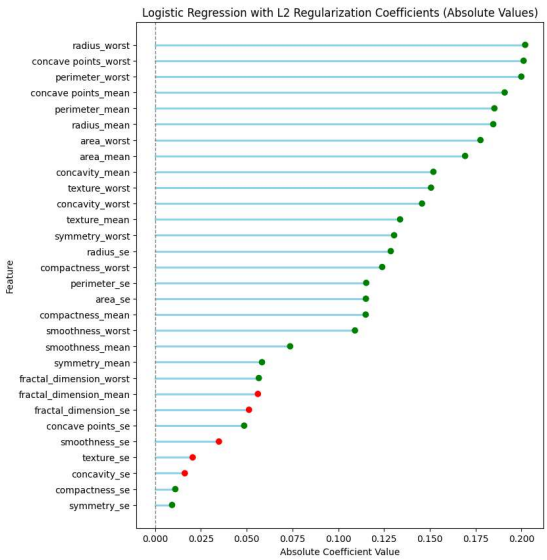
Finally, model performance was evaluated using confusion matrices and accuracy scores.

IV. Results

The malignancy of a cancer can be predicted with an accuracy of 97.5% using a logistic regression or a neural network. This is better than relying on a logistic regression with L1 regularization or L2 regularization, with respective accuracies of 94.3% and 94.0%, or a Random Forest, which has an accuracy of 94.0%. With those machine learning methods, we can predict whether a cancer is malignant or benign with a very good certainty knowing the characteristics of the cell nuclei.

Feature importance analysis highlights that certain clinical variables contribute significantly to classification performance. This reinforces the importance of interpretability when deploying predictive tools in healthcare contexts.





V. Conclusion

This project provides an illustrative application of supervised machine learning methods to clinical cancer classification. While the models achieve strong predictive performance, this work represents only a limited excerpt of my broader methodological capabilities.

A more comprehensive analysis would have required a deeper bias–variance study to better assess model generalization and potential overfitting. In a medical context, particular emphasis should also be placed on recall, especially to minimize false negatives. Failing to detect a malignant tumour carries significantly higher clinical risk than incorrectly classifying a benign case.

Despite these limitations, the implemented models demonstrate two important strengths. First, they provide reliable classification performance on structured clinical data. Second, beyond prediction alone, they allow the identification of key features associated with malignancy. Feature importance analysis and model coefficients highlight which clinical variables contribute most strongly to tumour classification, offering valuable interpretability.

In this sense, the project not only illustrates predictive modelling but also reflects a core objective of healthcare engineering: combining performance with interpretability to support informed clinical decision-making.